

1. Hospital Patient Billing System

```
import java.util.Scanner;

class HospitalBilling{
    int patientId;
    String patientName;
    String roomType;
    int days;
    double treatmentCharges;
    double roomCharges, totalBill;

    void readDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Patient ID: ");
        patientId = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Patient Name: ");
        patientName = sc.nextLine();

        System.out.print("Enter Room Type (General/Special): ");
        roomType = sc.nextLine();

        System.out.print("Enter Number of Days Admitted: ");
        days = sc.nextInt();

        System.out.print("Enter Treatment Charges: ");
        treatmentCharges = sc.nextDouble();
    }

    void calculateRoomCharges() {
        if (roomType.equalsIgnoreCase("General"))
            roomCharges = days * 1500;
        else
            roomCharges = days * 3000;
    }
}
```

```

void calculateBill() {
    totalBill = roomCharges + treatmentCharges;
}

void display() {
    System.out.println("\n----- Patient Bill -----");
    System.out.println("Patient ID: " + patientId);
    System.out.println("Patient Name: " + patientName);
    System.out.println("Room Charges: ₹" + roomCharges);
    System.out.println("Treatment Charges: ₹" + treatmentCharges);
    System.out.println("Total Bill: ₹" + totalBill);
}

public static void main(String[] args) {
    HospitalBilling h = new HospitalBilling();
    h.readDetails();
    h.calculateRoomCharges();
    h.calculateBill();
    h.display();
}
}

```

2. Airline Ticket Reservation System

```

import java.util.Scanner;

class AirlineReservation {
    int passengerId;
    String passengerName;
    String flightNumber;
    double ticketFare;
    int tickets;

    double totalCost, discount, gst, finalAmount;

    void readDetails() {

```

```
Scanner sc = new Scanner(System.in);

System.out.print("Enter Passenger ID: ");
passengerId = sc.nextInt();
sc.nextLine();

System.out.print("Enter Passenger Name: ");
passengerName = sc.nextLine();

System.out.print("Enter Flight Number: ");
flightNumber = sc.nextLine();

System.out.print("Enter Ticket Fare: ");
ticketFare = sc.nextDouble();

System.out.print("Enter Number of Tickets: ");
tickets = sc.nextInt();
}

void calculateCost() {
    totalCost = ticketFare * tickets;

    if (tickets > 5)
        discount = totalCost * 0.05;

    double amountAfterDiscount = totalCost - discount;

    gst = amountAfterDiscount * 0.12;
    finalAmount = amountAfterDiscount + gst;
}

void display() {
    System.out.println("\n----- Booking Details -----");
    System.out.println("Passenger ID: " + passengerId);
    System.out.println("Passenger Name: " + passengerName);
    System.out.println("Flight Number: " + flightNumber);
    System.out.println("Total Cost: ₹" + totalCost);
    System.out.println("Discount: ₹" + discount);
}
```

```
        System.out.println("GST: ₹" + gst);
        System.out.println("Final Amount: ₹" + finalAmount);
    }

    public static void main(String[] args) {
        AirlineReservation a = new AirlineReservation();
        a.readDetails();
        a.calculateCost();
        a.display();
    }
}
```

3. Vehicle Rental Management System

```
import java.util.Scanner;

class VehicleRental {
    int customerId;
    String customerName;
    String vehicleType;
    int rentalDays;
    int lateDays;

    double rentalCharges, penalty, totalBill;

    void readDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Customer ID: ");
        customerId = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Customer Name: ");
        customerName = sc.nextLine();

        System.out.print("Enter Vehicle Type (Car/Bike): ");
        vehicleType = sc.nextLine();
    }
}
```

```

        System.out.print("Enter Rental Days: ");
        rentalDays = sc.nextInt();

        System.out.print("Enter Late Return Days: ");
        lateDays = sc.nextInt();
    }

    void calculateCharges() {
        if (vehicleType.equalsIgnoreCase("Car"))
            rentalCharges = rentalDays * 2000;
        else
            rentalCharges = rentalDays * 500;

        penalty = lateDays * 200;
        totalBill = rentalCharges + penalty;
    }

    void display() {
        System.out.println("\n----- Rental Bill -----");
        System.out.println("Customer ID: " + customerId);
        System.out.println("Customer Name: " + customerName);
        System.out.println("Rental Charges: ₹" + rentalCharges);
        System.out.println("Penalty: ₹" + penalty);
        System.out.println("Total Bill: ₹" + totalBill);
    }

    public static void main(String[] args) {
        VehicleRental v = new VehicleRental();
        v.readDetails();
        v.calculateCharges();
        v.display();
    }
}

```

4. Manufacturing Production Monitoring System

```
import java.util.Scanner;

class ProductionMonitoring {
    int productId;
    String productName;
    int targetQty;
    int producedQty;

    double efficiency;

    void readDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Product ID: ");
        productId = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Product Name: ");
        productName = sc.nextLine();

        System.out.print("Enter Target Quantity: ");
        targetQty = sc.nextInt();

        System.out.print("Enter Produced Quantity: ");
        producedQty = sc.nextInt();
    }

    void calculateEfficiency() {
        efficiency = ((double) producedQty / targetQty) * 100;
    }

    void display() {
        System.out.println("\n----- Production Report -----");
        System.out.println("Product ID: " + productId);
        System.out.println("Product Name: " + productName);
        System.out.println("Efficiency: " + efficiency + "%");

        if (producedQty >= targetQty)
```

```
        System.out.println("Status: Achieved");
    else
        System.out.println("Status: Not Achieved");
}

public static void main(String[] args) {
    ProductionMonitoring p = new ProductionMonitoring();
    p.readDetails();
    p.calculateEfficiency();
    p.display();
}
}
```

5. Online Course Enrollment System

```
import java.util.Scanner;

class CourseEnrollment {
    int studentId;
    String studentName;
    String courseName;
    double courseFee;

    double registrationFee = 500;
    double scholarship = 0;
    double payableAmount;

    void readDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Student ID: ");
        studentId = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Student Name: ");
        studentName = sc.nextLine();
    }
}
```

```

        System.out.print("Enter Course Name: ");
        courseName = sc.nextLine();

        System.out.print("Enter Course Fee: ");
        courseFee = sc.nextDouble();
    }

    void calculateAmount() {
        if (courseFee > 20000)
            scholarship = courseFee * 0.15;

        payableAmount = courseFee - scholarship + registrationFee;
    }

    void display() {
        System.out.println("\n----- Enrollment Details -----");
        System.out.println("Student ID: " + studentId);
        System.out.println("Student Name: " + studentName);
        System.out.println("Course Name: " + courseName);
        System.out.println("Course Fee: ₹" + courseFee);
        System.out.println("Scholarship Discount: ₹" + scholarship);
        System.out.println("Registration Fee: ₹" + registrationFee);
        System.out.println("Payable Amount: ₹" + payableAmount);
    }

    public static void main(String[] args) {
        CourseEnrollment c = new CourseEnrollment();
        c.readDetails();
        c.calculateAmount();
        c.display();
    }
}

```